Document Investigation Dossier - case_tampered

Generated by DocForensics. Visual findings are indicators of possible alteration, not proof of fraud. Sections follow the investigation brief (7.1 to 7.6).

Case at a glance

Documents	Findings	High	Medium	Low
4	9	6	3	0

- 4 page(s) from 4 file(s); 4 recognised as a supported type, 0 not recognised.

- Text: 0 page(s) read exactly from the PDF text layer, 4 page(s) read by OCR.

- 23 field(s) extracted, 0 table(s) found.

- Visual forensics: pixel-level checks on 4 page(s), PDF structure checks on 0 generated PDF page(s).

- Cross-document: 8 field(s) compared across documents (2 agree, 3 differ, 3 in one document only, 0 could not be compared); 3 document pair(s) linked by shared identifiers.

- 9 finding(s) in total.

7.1 Document Overview

Document	Type	Source	Processing status	SHA-256 (first 16)
application_form	application_form (class. conf. 0.8; Application form 0.89)	application_form.jpg (JPG, 221341 bytes), page 1 of 1, 1000x1300 px	recognised as application form; read by OCR (tesseract, mean confidence 0.95); 6 field(s); pixel-level forensics ran	b65f9da544765c44
bank_statement	bank_statement (class. conf. 0.7; Bank statement 0.95)	bank_statement.jpg (JPG, 222854 bytes), page 1 of 1, 1000x1300 px	recognised as bank statement; read by OCR (tesseract, mean confidence 0.95); 6 field(s); pixel-level forensics ran	4805d814c8f5b122
id_card	id_card (class. conf. 0.75; ID card 0.97)	id_card.jpg (JPG, 112492 bytes), page 1 of 1, 1000x1300 px	recognised as id card; read by OCR (tesseract, mean confidence 0.949); 5 field(s); pixel-level forensics ran	c47d6e90089bbd98
salary_slip	salary_slip (class. conf. 0.79; Salary slip 0.89)	salary_slip.jpg (JPG, 114180 bytes), page 1 of 1, 1000x1300 px	recognised as salary slip; read by OCR (tesseract, mean confidence 0.952); 6 field(s); pixel-level forensics ran	7ef5677dede75362

7.2 Extracted Information

Field values with the confidence they were read with (1.00 for text read exactly from a PDF text layer, OCR word confidence for scans and photos) and their location in the original page: [x0, y0, x1, y1] in pixels of the page image at 200 dpi (divide by 2.78 for PDF points). The document structure follows.

application_form (application_form)

Field	Value	Read conf.	Location (page px)
name	Rohan Mehta	0.96	[433, 195, 612, 216]
dob	14/03/1994	0.96	[433, 281, 590, 304]
id_number	ABCDE1234F	0.91	[430, 366, 613, 386]
address	Flat 12, Sector 21, Noida UP 201301	0.96	[433, 450, 937, 475]
employer	Nimbus Analytics Pvt Ltd	0.96	[433, 535, 776, 562]
declared_income	85,000	0.96	[484, 621, 578, 645]

bank_statement (bank_statement)

Field	Value	Read conf.	Location (page px)
name	Rohan Mehra	0.96	[433, 195, 613, 216]
account_no	50100234567891	0.91	[432, 281, 677, 301]
period	01/10/2026 - 31/10/2026	0.94	[432, 366, 779, 389]
opening_balance	Rs. 120,000	0.96	[433, 451, 596, 475]
salary_credit	Rs. 58,000	0.96	[433, 536, 578, 560]
closing_balance	Rs. 146,500	0.96	[433, 621, 596, 645]

id_card (id_card)

Field	Value	Read conf.	Location (page px)
name	Rohan Mehta	0.96	[433, 195, 612, 216]
dob	14/03/1990	0.96	[434, 285, 590, 308]
id_number	ABCDE1284F	0.91	[430, 366, 613, 386]
address	Flat 12, Sector 21, Noida UP 201301	0.96	[433, 450, 937, 475]
issue_date	02/06/2019	0.96	[432, 536, 590, 559]

salary_slip (salary_slip)

Field	Value	Read conf.	Location (page px)
name	Rohan Mehta	0.96	[433, 195, 612, 216]
employer	Nimbus Analytics Pvt Ltd	0.95	[433, 280, 776, 307]
pay_month	October 2026	0.96	[432, 365, 620, 386]
gross	Rs. 95,000	0.95	[433, 451, 578, 475]
deductions	Rs. 37,000	0.96	[433, 536, 578, 560]
net_pay	Rs. 85,000	0.96	[432, 625, 579, 648]

Document structure: sections and tables

Sections are found from headings; tables from their header row. Values are read from the OCR word boxes; checks only use rows whose amounts could be read.

application_form (application_form)

Layout: label-value form. Hierarchy: page > 1 section(s) > 6 field(s), 0 table(s) (0 rows, 0 cells). Evidence: 6 labelled field(s); 0 label-free field(s); 1 heading(s).

Section	Title	Kind	Fields inside	Tables
S1	LOAN APPLICATION FORM	other	name, dob, id_number, address, employer, declared_income	-

bank_statement (bank_statement)

Layout: label-value form. Hierarchy: page > 1 section(s) > 6 field(s), 0 table(s) (0 rows, 0 cells). Evidence: 6 labelled field(s); 0 label-free field(s); 1 heading(s).

Section	Title	Kind	Fields inside	Tables
S1	BANK ACCOUNT STATEMENT	account_details	name, account_no, period, opening_balance, salary_credit, closing_balance	-

id_card (id_card)

Layout: label-value form. Hierarchy: page > 1 section(s) > 5 field(s), 0 table(s) (0 rows, 0 cells). Evidence: 5 labelled field(s); 0 label-free field(s); 1 heading(s).

Section	Title	Kind	Fields inside	Tables
S1	NATIONAL IDENTITY CARD	personal_details	name, dob, id_number, address, issue_date	-

salary_slip (salary_slip)

Layout: label-value form. Hierarchy: page > 1 section(s) > 6 field(s), 0 table(s) (0 rows, 0 cells). Evidence: 6 labelled field(s); 0 label-free field(s); 1 heading(s).

Section	Title	Kind	Fields inside	Tables
S1	SALARY SLIP	other	name, employer, pay_month, gross, deductions, net_pay	-

7.3 Contextual Analysis

How to read the status: potential alteration = an indicator that something may have been changed, not proof; inconsistency = a confirmed data conflict between values in the documents, which says nothing about intent; observation = a note, coverage gap or weak signal. Nothing here is a finding of fraud.

Identified field-level inconsistencies

Rules that must hold inside one document: arithmetic, totals and running balances, formats, checksums, date logic.

C-001 [HIGH | confidence high (0.98) | inconsistency] Salary slip arithmetic does not balance in salary_slip

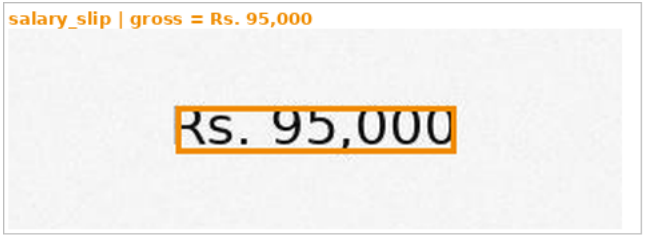
Explanation: Gross 95,000 - deductions 37,000 = 58,000, but Net Pay shows 85,000 (difference +27,000). Corroborated by visual evidence (V-002) on the same field.

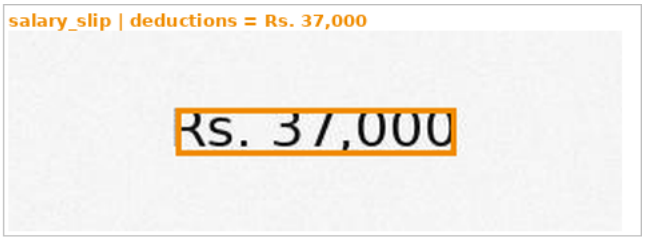
Evidence: salary_slip[7ef5677d].gross='Rs. 95,000'; salary_slip[7ef5677d].deductions='Rs. 37,000'; salary_slip[7ef5677d].net_pay='Rs. 85,000' *Method:* arithmetic check

Status: Confirmed data conflict: two values in the submitted documents contradict each other (this says nothing about intent)

Why it needs a look: Figures printed on the same page do not add up, so at least one of them is wrong or was changed after the document was produced.

Suggested check: Ask the issuer for the original or a recalculation and compare the figures with the payroll record.







Missing or duplicated information

Expected fields that were absent or unreadable, the same field printed twice with different values, and values that sit under the wrong label.

None identified.

Conflicting information across documents

Every field present in more than one document, with the value in each. 'agree' includes written-differently-but-compatible values (initials, omitted middle name, suffixes).

Field	Result	Values by document
Name	DIFFER	application_form: Rohan Mehta; bank_statement: Rohan Mehra; id_card: Rohan Mehta; salary_slip: Rohan Mehta
Date of birth	DIFFER	application_form: 14/03/1994; id_card: 14/03/1990
ID number	DIFFER	application_form: ABCDE1234F; id_card: ABCDE1284F
Address	agree	application_form: Flat 12, Sector 21, Noida UP 201301; id_card: Flat 12, Sector 21, Noida UP 201301
Employer	agree	application_form: Nimbus Analytics Pvt Ltd; salary_slip: Nimbus Analytics Pvt Ltd
Account number	one document only	bank_statement: 50100234567891
Monthly gross income	one document only	salary_slip: Rs. 95,000
Pay month	one document only	salary_slip: October 2026

Documents linked by shared identifiers: application_form - id_card (name); application_form - salary_slip (name); id_card - salary_slip (name)

Documents sharing no identifier with the others: bank_statement

Cross-document discrepancies

X-002 [HIGH | confidence high (0.95) | inconsistency] Date of birth mismatch: application_form vs id_card

Explanation: id_card: '14/03/1990' vs [application_form: '14/03/1994']. Values differ only slightly (similarity 90%): possible OCR slip, typo, transliteration or minor alteration. No majority: cannot tell which document is correct. Corroborated by visual evidence (V-001) on the same field.

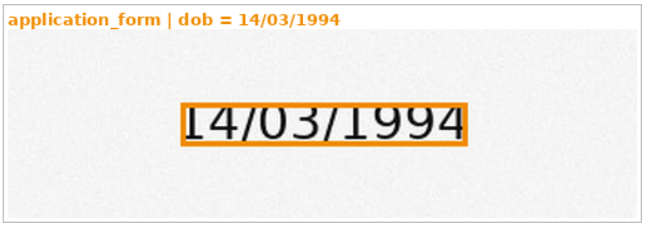
Evidence: id_card[c47d6e90].dob='14/03/1990'; application_form[b65f9da5].dob='14/03/1994' *Method:* cross-document dob comparison

Status: Confirmed data conflict: two values in the submitted documents contradict each other (this says nothing about intent)

Why it needs a look: Documents that describe the same person or application disagree; at most one of them can be right.

Suggested check: Obtain the originals of the documents named, find which one matches the issuer's record, and do not assume which is correct.





X-004 [HIGH | confidence high (0.95) | inconsistency] Salary slip net pay does not match bank payroll credit

Explanation: Net pay on salary_slip is 85,000, but the payroll credit in the bank statement (bank_statement) is 58,000 (difference +27,000). Either the payslip or the bank record is wrong; this is a data inconsistency to verify with the employer and the bank, not proof of fraud. Corroborated by visual evidence (V-002) on the same field.

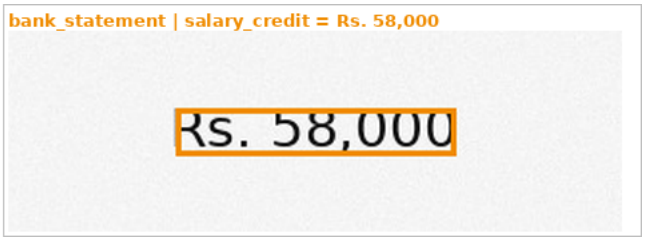
Evidence: salary_slip[7ef5677d].net_pay='Rs. 85,000'; bank_statement[4805d814].salary_credit='Rs. 58,000' *Method:* income reconciliation

Status: Confirmed data conflict: two values in the submitted documents contradict each other (this says nothing about intent)

Why it needs a look: Records that should agree (payslip, bank statement, application, case reference) show different values.

Suggested check: Obtain the bank's own statement and the employer's payroll or confirmation record and see which side matches the issuer's data.





X-003 [HIGH | confidence high (0.90) | inconsistency] ID number mismatch: application_form vs id_card

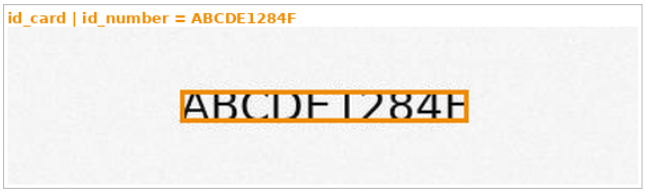
Explanation: id_card: 'ABCDE1284F' vs [application_form: 'ABCDE1234F']. Values differ only slightly (similarity 90%): possible OCR slip, typo, transliteration or minor alteration. No majority: cannot tell which document is correct.

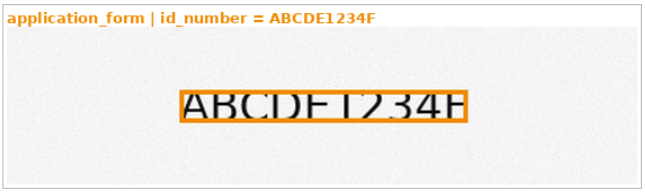
Evidence: id_card[c47d6e90].id_number='ABCDE1284F'; application_form[b65f9da5].id_number='ABCDE1234F' *Method:* cross-document id_number comparison

Status: Confirmed data conflict: two values in the submitted documents contradict each other (this says nothing about intent)

Why it needs a look: Documents that describe the same person or application disagree; at most one of them can be right.

Suggested check: Obtain the originals of the documents named, find which one matches the issuer's record, and do not assume which is correct.





X-001 [MEDIUM | confidence high (0.80) | inconsistency] Applicant name in bank_statement disagrees with 3 other document(s)

Explanation: bank_statement: 'Rohan Mehra' vs [application_form: 'Rohan Mehta', id_card: 'Rohan Mehta', salary_slip: 'Rohan Mehta']. Values differ only slightly (similarity 90%): possible OCR slip, typo, transliteration or minor alteration.

Evidence: bank_statement[4805d814].name='Rohan Mehra'; application_form[b65f9da5].name='Rohan Mehta'; id_card[c47d6e90].name='Rohan Mehta'; salary_slip[7ef5677d].name='Rohan Mehta' *Method:* cross-document name comparison

Status: Confirmed data conflict: two values in the submitted documents contradict each other (this says nothing about intent)

Why it needs a look: Documents that describe the same person or application disagree; at most one of them can be right.

Suggested check: Obtain the originals of the documents named, find which one matches the issuer's record, and do not assume which is correct.









X-005 [MEDIUM | confidence medium (0.70) | inconsistency] Declared income exceeds income shown in bank_statement

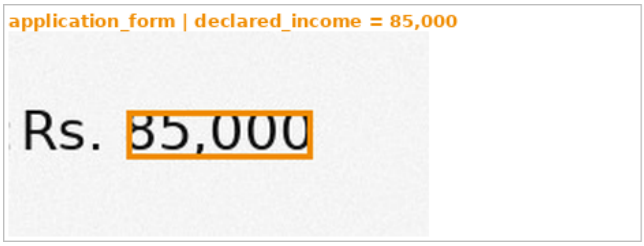
Explanation: Declared monthly income 85,000 vs the highest amount in the documents, bank_statement salary credit 58,000 (47% higher). No payslip gross is available, so the declared figure may be gross while the documents show net pay or credits; verify which basis was declared.

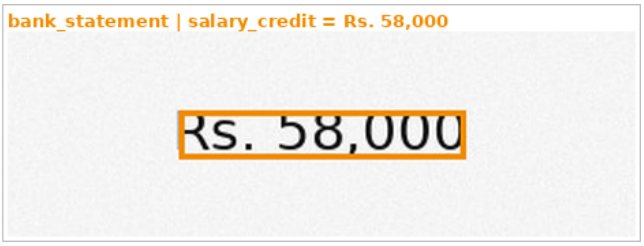
Evidence: application_form[b65f9da5].declared_income='85,000'; bank_statement[4805d814].salary_credit='Rs. 58,000' *Method:* income reconciliation

Status: Confirmed data conflict: two values in the submitted documents contradict each other (this says nothing about intent)

Why it needs a look: Records that should agree (payslip, bank statement, application, case reference) show different values.

Suggested check: Obtain the bank's own statement and the employer's payroll or confirmation record and see which side matches the issuer's data.





R-001 [MEDIUM | confidence medium (0.60) | observed] bank_statement shares no identifier with the other documents

Explanation: None of the identifiers that tie documents to one person or application (name, date of birth, ID number, employee ID, account number, case reference) agree between bank_statement and any other document of this case, although the other documents are linked to each other. It may belong to a different applicant or application; verify before relying on it.

Evidence: bank_statement[4805d814] (whole page) *Method:* document linkage

Status: Observation: a note, coverage gap or weak signal, not an accusation

Why it needs a look: Documents that describe the same person or application disagree; at most one of them can be right.

Suggested check: Obtain the originals of the documents named, find which one matches the issuer's record, and do not assume which is correct.

No region to highlight: this finding concerns the whole page or a value that could not be located.

7.4 Visual Forensic Findings

Pixel-level and PDF-structure indicators that a region may differ from the rest of the page. They show where to look; they do not prove alteration. Each group says what was checked, so a group without findings states what was examined and found nothing.

Suspicious regions and potential alterations

V-001 [HIGH | confidence high (0.98) | potential manipulation] Possible alteration of 'Date of Birth' in id_card

Explanation: The value region of 'Date of Birth' ('14/03/1990') differs statistically from the other fields of the same page in: bglum (z=+3.0), inkdark (z=+11.4). Such a pattern is typical of text that was edited or pasted after the page was first scanned/compressed. This is an indicator, not proof; verify against the original document.

Evidence: id_card[c47d6e90].dob='14/03/1990' *Method:* peer-relative ELA/noise/background/stroke analysis

Status: Potential alteration: an indicator that something may have been changed, not proof

Why it needs a look: Pixels in this region behave differently from the rest of the page (compression, grain, tone, ink or typography). Edits can leave such traces, so can scanning.

Suggested check: Look at the region at full resolution on the original capture, repeat the capture if possible, and compare with the issuer's copy.



V-002 [HIGH | confidence high (0.98) | potential manipulation] Possible alteration of 'Net Pay' in salary_slip

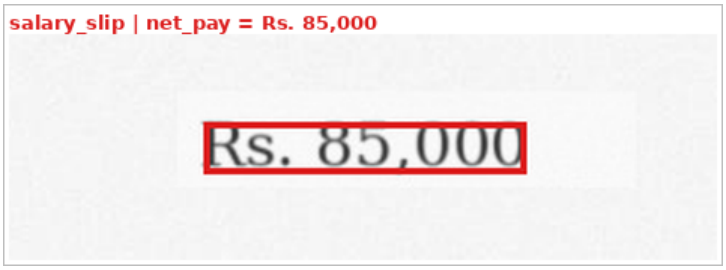
Explanation: The value region of 'Net Pay' ('Rs. 85,000') differs statistically from the other fields of the same page in: noise (z=-4.5), bglum (z=+4.5), inkdark (z=+10.5). Such a pattern is typical of text that was edited or pasted after the page was first scanned/compressed. This is an indicator, not proof; verify against the original document.

Evidence: salary_slip[7ef5677d].net_pay='Rs. 85,000' *Method:* peer-relative ELA/noise/background/stroke analysis

Status: Potential alteration: an indicator that something may have been changed, not proof

Why it needs a look: Pixels in this region behave differently from the rest of the page (compression, grain, tone, ink or typography). Edits can leave such traces, so can scanning.

Suggested check: Look at the region at full resolution on the original capture, repeat the capture if possible, and compare with the issuer's copy.



Font, alignment and structural inconsistencies

Checked: application_form: typography run on 5 fields | bank_statement: typography run on 6 fields | id_card: typography run on 5 fields | salary_slip: typography run on 6 fields

None identified.

Image manipulation observations

Checked: application_form: pixel forensics run; JPEG grid run on 266 tiles; dominant grid 100%; photo not an ID card | bank_statement: pixel forensics run; JPEG grid run on 258 tiles; dominant grid 100%; photo not an ID card | id_card: pixel forensics run; JPEG grid run on 298 tiles; dominant grid 100%; photo no photo-shaped continuous-tone region located (busy background, no photo, or photo too small) | salary_slip: pixel forensics run; JPEG grid run on 300 tiles; dominant grid 100%; photo not an ID card

None identified.

Signature-related observations

Checked: application_form: no signature label on the page | bank_statement: no signature label on the page | id_card: no signature label on the page | salary_slip: no signature label on the page

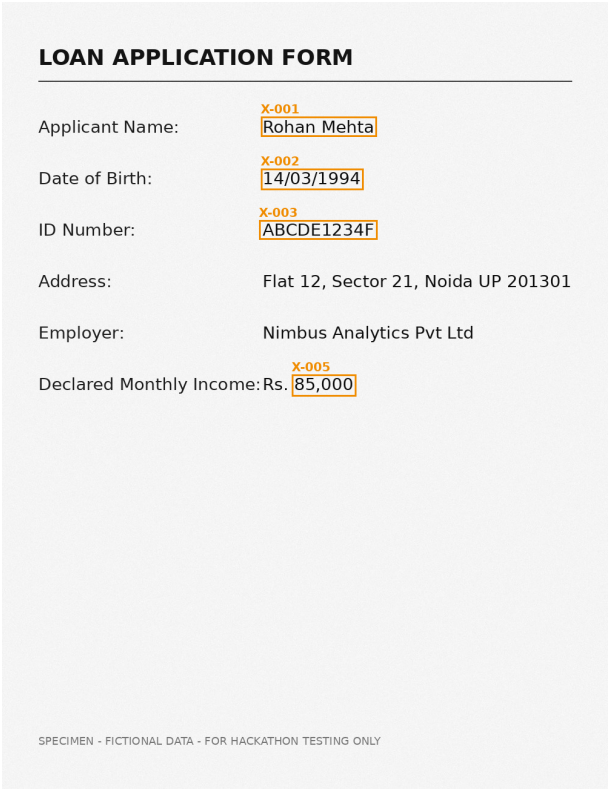
None identified.

7.5 Evidence and Traceability

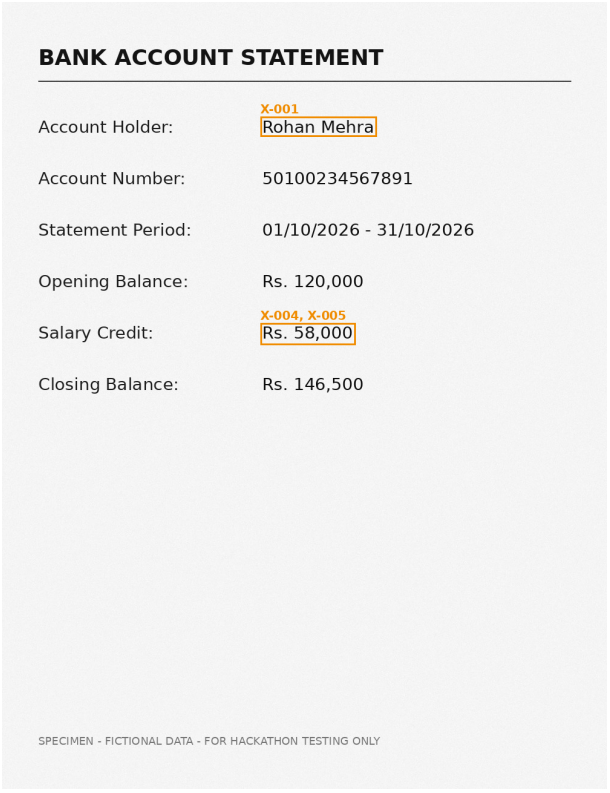
Each finding is tied to its original documents (file hash), the extracted fields and values it rests on, and the region it points to (box as in 7.2). Zoomed crops are printed with each finding in 7.3 and 7.4; the highlighted pages follow the table.

ID	Documents [SHA-256 first 8]	Associated extracted information	Region [x0,y0,x1,y1]	Status	Confidence
V-001	id_card[c47d6e90]	id_card.dob=14/03/1990	[434, 285, 590, 308]	potential manipulation	high (0.98)
V-002	salary_slip[7ef5677d]	salary_slip.net_pay=Rs. 85,000	[432, 625, 579, 648]	potential manipulation	high (0.98)
C-001	salary_slip[7ef5677d]	salary_slip.gross=Rs. 95,000; salary_slip.deductions=Rs. 37,000; salary_slip.net_pay=Rs. 85,000	[433, 451, 578, 475]; [433, 536, 578, 560]; [432, 625, 579, 648]	inconsistency	high (0.98)
X-002	application_form[b65f9da5], id_card[c47d6e90]	id_card.dob=14/03/1990; application_form.dob=14/03/1994	[434, 285, 590, 308]; [433, 281, 590, 304]	inconsistency	high (0.95)
X-004	bank_statement[4805d814], salary_slip[7ef5677d]	salary_slip.net_pay=Rs. 85,000; bank_statement.salary_credit=Rs. 58,000	[432, 625, 579, 648]; [433, 536, 578, 560]	inconsistency	high (0.95)
X-003	application_form[b65f9da5], id_card[c47d6e90]	id_card.id_number=ABCDE1284F; application_form.id_number=ABCDE1234F	[430, 366, 613, 386]; [430, 366, 613, 386]	inconsistency	high (0.90)
X-001	application_form[b65f9da5], bank_statement[4805d814], id_card[c47d6e90], salary_slip[7ef5677d]	bank_statement.name=Rohan Mehra; application_form.name=Rohan Mehta; id_card.name=Rohan Mehta; salary_slip.name=Rohan Mehta	[433, 195, 613, 216]; [433, 195, 612, 216]; [433, 195, 612, 216]; [433, 195, 612, 216]	inconsistency	high (0.80)
X-005	application_form[b65f9da5], bank_statement[4805d814]	application_form.declared_income=85,000; bank_statement.salary_credit=Rs. 58,000	[484, 621, 578, 645]; [433, 536, 578, 560]	inconsistency	medium (0.70)
R-001	bank_statement[4805d814]	(whole page)	-	observed	medium (0.60)

Highlighted regions (boxes are labelled with finding IDs; colours: red potential alteration, orange inconsistency, blue observation)



application_form



bank_statement

NATIONAL IDENTITY CARD

Name: Rohan Mehta

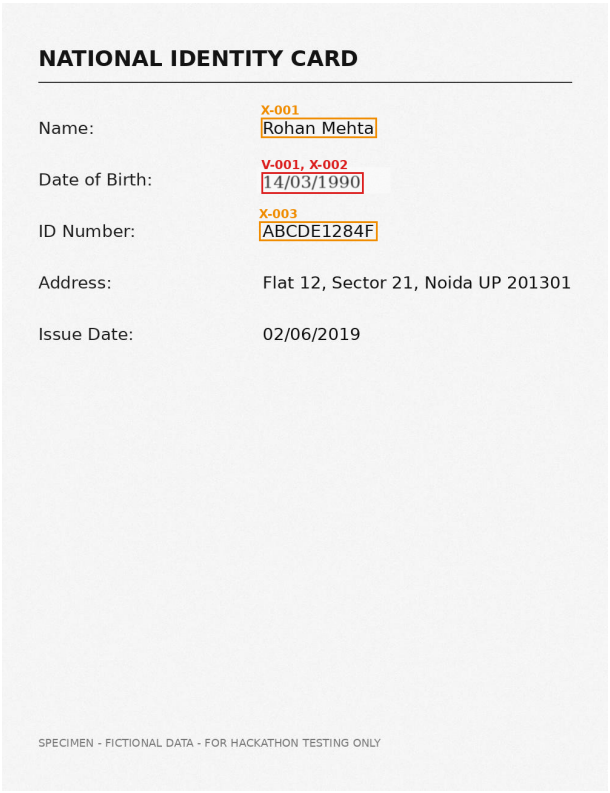
Date of Birth: 14/03/1990

ID Number: ABCDE1284F

Address: Flat 12, Sector 21, Noida UP 201301

Issue Date: 02/06/2019

SPECIMEN - FICTIONAL DATA - FOR HACKATHON TESTING ONLY



id_card

SALARY SLIP

Employee Name: Rohan Mehta

Employer: Nimbus Analytics Pvt Ltd

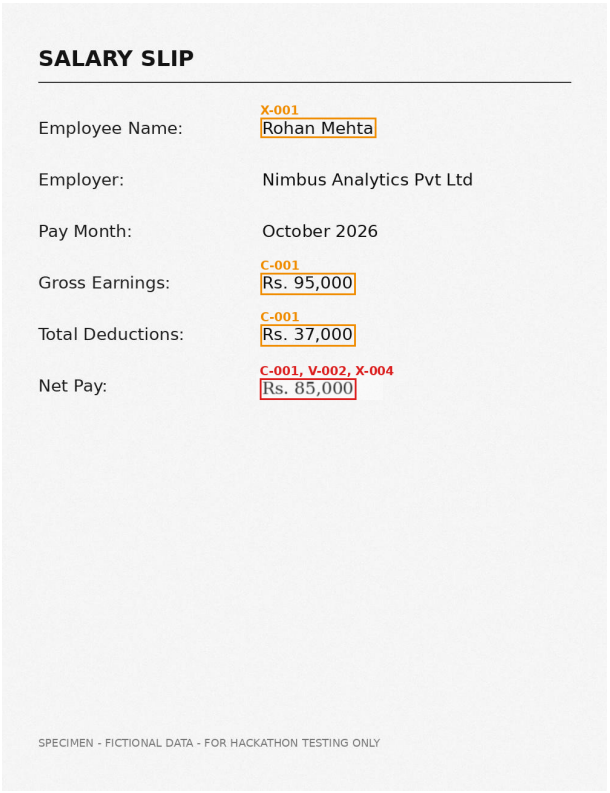
Pay Month: October 2026

Gross Earnings: Rs. 95,000

Total Deductions: Rs. 37,000

Net Pay: Rs. 85,000

SPECIMEN - FICTIONAL DATA - FOR HACKATHON TESTING ONLY



salary_slip

7.6 Investigation Summary

Case case_tampered: 4 documents analysed; 9 finding(s) about the documents (6 high, 3 medium, 0 low). Most significant: V-001 Possible alteration of 'Date of Birth' in id_card; V-002 Possible alteration of 'Net Pay' in salary_slip; C-001 Salary slip arithmetic does not balance in salary_slip; X-002 Date of birth mismatch: application_form vs id_card. Visual findings are indicators of possible alteration, not proof; confirm against originals. Pixel-level visual forensics ran on 4 of 4 recognised page(s).

Summary source: template

Identified anomalies

ID	Anomaly	Severity	Status	Confidence	Documents
V-001	Possible alteration of 'Date of Birth' in id_card	high	potential manipulation	high (0.98)	id_card
V-002	Possible alteration of 'Net Pay' in salary_slip	high	potential manipulation	high (0.98)	salary_slip
C-001	Salary slip arithmetic does not balance in salary_slip	high	inconsistency	high (0.98)	salary_slip
X-002	Date of birth mismatch: application_form vs id_card	high	inconsistency	high (0.95)	application_form, id_card
X-004	Salary slip net pay does not match bank payroll credit	high	inconsistency	high (0.95)	bank_statement, salary_slip
X-003	ID number mismatch: application_form vs id_card	high	inconsistency	high (0.90)	application_form, id_card
X-001	Applicant name in bank_statement disagrees with 3 other document(s)	medium	inconsistency	high (0.80)	application_form, bank_statement, id_card, salary_slip
X-005	Declared income exceeds income shown in bank_statement	medium	inconsistency	medium (0.70)	application_form, bank_statement
R-001	bank_statement shares no identifier with the other documents	medium	observed	medium (0.60)	bank_statement

Supporting evidence

Every anomaly above rests on the references, zoomed crops and highlighted regions in 7.3 to 7.5, and on the hash-chained audit log (appendix C), which records the references, the SHA-256 of the source file and of the evidence image, and the processing events that produced them.

Potential explanations

Possible, non-exclusive explanations to rule in or out when checking the originals. They are hypotheses, not conclusions.

ID	Possible explanations
V-001	Scanning, compression or lighting artefacts · A value overwritten or pasted in after printing · Different fonts or printers used legitimately on the same page
V-002	Scanning, compression or lighting artefacts · A value overwritten or pasted in after printing · Different fonts or printers used legitimately on the same page
C-001	A typing or calculation error when the document was produced · One figure edited after the document was produced · A different version of the statement mixed with this one
X-002	A typing error in one of the documents · The documents were produced at different times or are different versions · A value in one document was changed · A document of another person was mixed in
X-004	Timing: the payslip was corrected or reissued after the bank credit · Net pay shown before an adjustment that the bank reflects · A figure on the payslip or in the bank record was changed
X-003	A typing error in one of the documents · The documents were produced at different times or are different versions · A value in one document was changed · A document of another person was mixed in
X-001	A typing error in one of the documents · The documents were produced at different times or are different versions · A value in one document was changed · A document of another person was mixed in
X-005	Timing: the payslip was corrected or reissued after the bank credit · Net pay shown before an adjustment that the bank reflects · A figure on the payslip or in the bank record was changed

ID	Possible explanations
R-001	A document from another application was added by mistake · A clerical error in the reference · A reference altered to attach the document to this case

Areas requiring additional verification

Area	Why it needs verification
Fields present in only one document	Account number, Monthly gross income, Pay month. They could not be cross-checked; confirm them against an independent source.
bank_statement: shares no identifier with the other documents	Confirm that it belongs to this application.
Issuer verification	No document was checked against the issuing organisation or a registry. Confirm identity, employment and account details with the issuers.
Original documents	Findings come from the submitted copies only. Compare every flagged value and region with the originals before any decision.

Investigation limitations

Peer-relative statistics need at least four extracted fields per page. Heavily re-compressed or re-scanned documents reduce visual-forensic sensitivity. OCR errors can create false content mismatches. Skilled forgeries that match font, ink, tone and keep all numbers consistent are not detected by pixel forensics. Generated PDFs get structure checks only. No document was verified with its issuer. All flagged items should be verified against the original documents and the issuing authority.

Confidence levels: high (0.80 and above), medium (0.60 to 0.79), low (below 0.60). The number is the system's own estimate of how reliable the specific check is on this evidence (for example an exact arithmetic mismatch scores high, a weak pixel signal scores low). It is not a probability that a document is forged, and it does not say which of two conflicting documents is correct.

Appendix A. Capabilities and limitations of this system

Capability	Status
Ingestion: images, PDFs and ZIP bundles (one zip = one case); multi-document cases	Implemented
Page normalisation: deskew (tilt up to +/-8 deg) and lighting flattening for phone photos	Implemented; perspective correction planned
OCR + field extraction with field-level confidence and bounding boxes	Implemented (Tesseract + optional RapidOCR; printed text, label:value layouts)
Semantic label matching (embeddings) for field labels outside the dictionary	Implemented (needs embedding model; falls back to fuzzy dictionary)
Document classification: embeddings + keywords, unsupported-class veto, structural gate (>=3 expected fields)	Implemented (keyword + structure fallback when no model)
Visual forensics: ELA, grain/noise, paper tone, ink darkness (peer-relative, lighting-corrected)	Implemented
Contextual checks: arithmetic, formats, date logic, completeness	Implemented
Cross-document entity matching (odd-one-out) and income reconciliation	Implemented
Section detection; transaction and earnings/deductions tables with running-balance, page-continuity, total and component-sum checks	Implemented (bank-statement and salary-slip layouts)
Evidence mapping (annotated regions), audit trail, reviewer UI, PDF/JSON dossier	Implemented
Handwriting recognition	Unsupported
Typography / spacing / alignment checks, JPEG block-grid consistency (inserted or replaced content), photo-region check on ID cards, signature detection with placement and background checks	Implemented (heuristic; see limitations)
Born-digital PDFs: exact text layer (no OCR), font consistency of amounts, bordered patches over values, covered or hidden text, file history (revisions, dates, producer)	Implemented (observations about file structure, not proof)
Duplicated (copy-move) image regions on scans and photos; image metadata (editing software, modification after capture); scanned-PDF file history	Implemented (observations; copies recompressed below about JPEG quality 92 after pasting can be missed)
Signature authenticity, photo identity matching	Planned
Layout-aware ML models (LayoutLM etc.), tables in other layouts, multilingual OCR	Planned

Appendix B. Standards and reference frameworks (no compliance claim)

This system draws on the frameworks below as reference points. It does not claim compliance or certification with any of them; each row says what we do and what we do not do.

Framework	What we do	What we do not do
ISO/IEC 27037 (identification, collection, acquisition and preservation of digital evidence)	Each source file is hashed (SHA-256) at ingest; originals are never modified (analysis uses derived copies, recorded in the log); every step is written to a hash-chained audit log with archived run history.	No acquisition or custody control before upload: the system only sees the copy it is given. No write protection or collection records for the original device or system.
ISO/IEC 27041 (suitability and adequacy of investigative methods)	Methods are documented (ARCHITECTURE.md), tested on a synthetic benchmark and an organiser test pack with an answer key, and limitations are stated openly (skilled forgeries, tuning on synthetic data).	No formal validation study, no measured error rates on real documents, no independent testing.
ISO/IEC 27042 (analysis and interpretation of digital evidence)	Findings separate a data inconsistency, a potential alteration and an observation; each has a confidence level, possible explanations and a suggested check; nothing is called fraud; the pipeline is deterministic and repeatable.	Confidence levels are the system's own estimates, not calibrated probabilities; no formal competence or review process around the analyst.
NIST SP 800-86 (forensic techniques in incident response)	The pipeline follows its collection, examination, analysis and reporting phases: ingest and hashing; reading, extraction, structure and forensic checks; contextual and cross-document analysis; the dossier.	It is written for incident response on live systems; volatile data, network and system sources are out of scope here.

Framework	What we do	What we do not do
OWASP Machine Learning Security Top 10	No training is done and no model is hosted, so model theft, poisoning, inversion and membership inference do not apply. Supply chain: the downloaded embedding model is checked against recorded SHA-256 values and the result is logged. Output integrity: hash-chained audit and review logs. The optional Claude summary is off by default.	Input manipulation (documents crafted to mislead OCR or classification) is only partly mitigated (veto rule, minimum expected fields) and not tested adversarially. Recorded model hashes were taken from our own first download, not from a publisher signature. RapidOCR models are bundled in their package and not separately pinned. When the optional summary is on, finding text (document values) is sent to an external API.

Appendix C. Audit trail and investigation history

Hash-chained event log of run 20261105T054413-dcf1ee: 56 events, chain **verified intact** (checked in memory and as written to disk). Digest of the final event (SHA-256): 432981265a7f117da7838bb584450f6145101acadbd46769a19b3e2f8951bc51

Each finding record in the log carries its references, the SHA-256 of the source file and of the evidence crop, and the hashes of the ingest, OCR/text-layer, extraction and field-report events that produced it. References are shown as document[first 8 characters of the file's SHA-256]. Reviewer decisions are kept in review_log.jsonl (separate hash chain, anchored to this run's digest).

Earlier runs of this case: none (first run)

Time	Event	Detail
05:44:13	case_start	{'run': '20261105T054413-dcf1ee', 'case_id': 'case_tampered', 'n_inputs': 4, 'run_id': '20261105T054413-dcf1ee
05:44:14	embedding_model	{'run': '20261105T054413-dcf1ee', 'backend': 'fastembed:BAAI/bge-small-en-v1.5', 'active': True, 'integrity':
05:44:14	deskew	{'run': '20261105T054413-dcf1ee', 'doc_id': 'application_form', 'estimated_angle': -0.0, 'gain': 1.0, 'applied
05:44:14	ingest	{'run': '20261105T054413-dcf1ee', 'doc_id': 'application_form', 'source': 'application_form.jpg', 'sha256': 'b
05:44:14	deskew	{'run': '20261105T054413-dcf1ee', 'doc_id': 'bank_statement', 'estimated_angle': -0.0, 'gain': 1.0, 'applied':
05:44:14	ingest	{'run': '20261105T054413-dcf1ee', 'doc_id': 'bank_statement', 'source': 'bank_statement.jpg', 'sha256': '4805d
05:44:14	deskew	{'run': '20261105T054413-dcf1ee', 'doc_id': 'id_card', 'estimated_angle': -0.0, 'gain': 1.0, 'applied': False}
05:44:15	ingest	{'run': '20261105T054413-dcf1ee', 'doc_id': 'id_card', 'source': 'id_card.jpg', 'sha256': 'c47d6e90089bbd980f1
05:44:15	deskew	{'run': '20261105T054413-dcf1ee', 'doc_id': 'salary_slip', 'estimated_angle': -0.0, 'gain': 1.0, 'applied': Fa
05:44:15	ingest	{'run': '20261105T054413-dcf1ee', 'doc_id': 'salary_slip', 'source': 'salary_slip.jpg', 'sha256': '7ef5677dede
05:44:16	ocr	{'run': '20261105T054413-dcf1ee', 'doc_id': 'application_form', 'engine': 'tesseract', 'n_words': 40, 'mean_co
05:44:16	extract	{'run': '20261105T054413-dcf1ee', 'doc_id': 'application_form', 'fields': ['name', 'dob', 'id_number', 'addres
05:44:17	classify	{'run': '20261105T054413-dcf1ee', 'doc_id': 'application_form', 'doc_type': 'application_form', 'best_guess':
05:44:18	ocr	{'run': '20261105T054413-dcf1ee', 'doc_id': 'bank_statement', 'engine': 'tesseract', 'n_words': 36, 'mean_conf
05:44:18	extract	{'run': '20261105T054413-dcf1ee', 'doc_id': 'bank_statement', 'fields': ['name', 'account_no', 'period', 'open
05:44:18	classify	{'run': '20261105T054413-dcf1ee', 'doc_id': 'bank_statement', 'doc_type': 'bank_statement', 'best_guess': 'ban
05:44:19	ocr	{'run': '20261105T054413-dcf1ee', 'doc_id': 'id_card', 'engine': 'tesseract', 'n_words': 33, 'mean_conf': 0.94
05:44:19	extract	{'run': '20261105T054413-dcf1ee', 'doc_id': 'id_card', 'fields': ['name', 'dob', 'id_number', 'address', 'issu

Time	Event	Detail
05:44:19	classify	{'run': '20261105T054413-dcf1ee', 'doc_id': 'id_card', 'doc_type': 'id_card', 'best_guess': 'id_card', 'keywor
05:44:20	ocr	{'run': '20261105T054413-dcf1ee', 'doc_id': 'salary_slip', 'engine': 'tesseract', 'n_words': 36, 'mean_conf':
05:44:20	extract	{'run': '20261105T054413-dcf1ee', 'doc_id': 'salary_slip', 'fields': ['name', 'employer', 'pay_month', 'gross'
05:44:20	classify	{'run': '20261105T054413-dcf1ee', 'doc_id': 'salary_slip', 'doc_type': 'salary_slip', 'best_guess': 'salary_sl
05:44:20	categorise	{'run': '20261105T054413-dcf1ee', 'file': 'application_form.jpg', 'label': 'Application form', 'confidence': 0
05:44:20	categorise	{'run': '20261105T054413-dcf1ee', 'file': 'bank_statement.jpg', 'label': 'Bank statement', 'confidence': 0.955
05:44:20	categorise	{'run': '20261105T054413-dcf1ee', 'file': 'id_card.jpg', 'label': 'ID card', 'confidence': 0.973, 'pages_agree
05:44:20	categorise	{'run': '20261105T054413-dcf1ee', 'file': 'salary_slip.jpg', 'label': 'Salary slip', 'confidence': 0.891, 'pag
05:44:20	structure	{'run': '20261105T054413-dcf1ee', 'doc_id': 'application_form', 'sections': [('LOAN APPLICATION FORM', 'other'
05:44:20	structure	{'run': '20261105T054413-dcf1ee', 'doc_id': 'bank_statement', 'sections': [('BANK ACCOUNT STATEMENT', 'account
05:44:20	structure	{'run': '20261105T054413-dcf1ee', 'doc_id': 'id_card', 'sections': [('NATIONAL IDENTITY CARD', 'personal_detai
05:44:20	structure	{'run': '20261105T054413-dcf1ee', 'doc_id': 'salary_slip', 'sections': [('SALARY SLIP', 'other')], 'tables': [
05:44:20	forensics	{'run': '20261105T054413-dcf1ee', 'doc_id': 'application_form', 'fields_tested': 6, 'flagged': []}
05:44:20	forensics	{'run': '20261105T054413-dcf1ee', 'doc_id': 'bank_statement', 'fields_tested': 6, 'flagged': []}
05:44:21	forensics	{'run': '20261105T054413-dcf1ee', 'doc_id': 'id_card', 'fields_tested': 5, 'flagged': ['dob']}
05:44:21	forensics	{'run': '20261105T054413-dcf1ee', 'doc_id': 'salary_slip', 'fields_tested': 6, 'flagged': ['net_pay']}
05:44:21	typography	{'run': '20261105T054413-dcf1ee', 'doc_id': 'application_form', 'fields': 5, 'flagged': []}
05:44:21	visual_ext	{'run': '20261105T054413-dcf1ee', 'doc_id': 'application_form', 'typography': 'run on 5 fields', 'grid': 'run
05:44:22	typography	{'run': '20261105T054413-dcf1ee', 'doc_id': 'bank_statement', 'fields': 6, 'flagged': []}
05:44:22	visual_ext	{'run': '20261105T054413-dcf1ee', 'doc_id': 'bank_statement', 'typography': 'run on 6 fields', 'grid': 'run on
05:44:23	typography	{'run': '20261105T054413-dcf1ee', 'doc_id': 'id_card', 'fields': 5, 'flagged': []}
05:44:23	visual_ext	{'run': '20261105T054413-dcf1ee', 'doc_id': 'id_card', 'typography': 'run on 5 fields', 'grid': 'run on 298 ti
05:44:24	typography	{'run': '20261105T054413-dcf1ee', 'doc_id': 'salary_slip', 'fields': 6, 'flagged': []}
05:44:24	visual_ext	{'run': '20261105T054413-dcf1ee', 'doc_id': 'salary_slip', 'typography': 'run on 6 fields', 'grid': 'run on 30
05:44:24	fields_final	{'run': '20261105T054413-dcf1ee', 'doc_id': 'application_form', 'doc_type': 'application_form', 'source_sha256
05:44:24	fields_final	{'run': '20261105T054413-dcf1ee', 'doc_id': 'bank_statement', 'doc_type': 'bank_statement', 'source_sha256': '
05:44:24	fields_final	{'run': '20261105T054413-dcf1ee', 'doc_id': 'id_card', 'doc_type': 'id_card', 'source_sha256': 'c47d6e90089bbd
05:44:24	fields_final	{'run': '20261105T054413-dcf1ee', 'doc_id': 'salary_slip', 'doc_type': 'salary_slip', 'source_sha256': '7ef567
05:44:25	finding	{'run': '20261105T054413-dcf1ee', 'fid': 'V-001', 'kind': 'visual', 'severity': 'high', 'confidence': 0.98, 's
05:44:25	finding	{'run': '20261105T054413-dcf1ee', 'fid': 'V-002', 'kind': 'visual', 'severity': 'high', 'confidence': 0.98, 's
05:44:25	finding	{'run': '20261105T054413-dcf1ee', 'fid': 'C-001', 'kind': 'intra', 'severity': 'high', 'confidence': 0.98, 'st

Time	Event	Detail
05:44:25	finding	{'run': '20261105T054413-dcf1ee', 'fid': 'X-001', 'kind': 'cross', 'severity': 'medium', 'confidence': 0.8, 's
05:44:25	finding	{'run': '20261105T054413-dcf1ee', 'fid': 'X-002', 'kind': 'cross', 'severity': 'high', 'confidence': 0.95, 'st
05:44:25	finding	{'run': '20261105T054413-dcf1ee', 'fid': 'X-003', 'kind': 'cross', 'severity': 'high', 'confidence': 0.9, 'sta
05:44:25	finding	{'run': '20261105T054413-dcf1ee', 'fid': 'X-004', 'kind': 'cross', 'severity': 'high', 'confidence': 0.95, 'st
05:44:25	finding	{'run': '20261105T054413-dcf1ee', 'fid': 'X-005', 'kind': 'cross', 'severity': 'medium', 'confidence': 0.7, 's
05:44:25	finding	{'run': '20261105T054413-dcf1ee', 'fid': 'R-001', 'kind': 'cross', 'severity': 'medium', 'confidence': 0.6, 's
05:44:26	dossier	{'run': '20261105T054413-dcf1ee', 'case_id': 'case_tampered', 'n_findings': 9, 'summary_source': 'template'}